

ONT Root And Subname Sovereignty Paths

Preliminary scale model: roots are the direct Bitcoin-sovereign primitive; subnames are the scalable namespace path, with optional hardening for higher-assurance cases.

Draft framing
for discussion

Roots require Bitcoin footprint and maturity, but are the cleanest "just own a name" path.

Subnames can be issued without a new UTXO if the root grants ownership to a subname key.

Finality can come from grant acceptance, checkpointing, and a non-revocation rule.

Example	What it is	Requires new UTXO?	Requires maturity?	Useful when?	Self-sovereign / final when?	Main caveat
bitcoin	Root name won through permissionless on-chain auction.	Yes	Yes Example: 1 year.	Day 1. Owner controls it, but bond continuity matters.	After root maturity. Owner-key authority can survive bond release.	Highest assurance, but consumes Bitcoin blockspace and one immature bond UTXO.
alice@bitcoin after bitcoin is mature	Subname issued by a mature root to Alice's owner key.	No	No No new subname maturity.	Seconds to minutes after root grant and Alice acceptance.	After checkpoint plus finality. Root cannot overwrite if grant is non-revocable.	Not root-equivalent. Validity depends on the root grant and proof history.
alice@bitcoin before bitcoin is mature	Subname issued while the root is still in its bond-maturity window.	No	Inherits root risk	Seconds to minutes after accepted grant.	After grant finality and root maturity.	If the root bond fails before maturity, dependent subnames may be affected.
agent7@alice	Scalable identity for an agent, app, device, department, or key.	No	No If root is mature.	Immediately after accepted grant, subject to resolver availability.	After checkpoint plus finality under the root's published policy.	Good for scale, but root policy and proof availability must be explicit.
treasury@acme	High-assurance subname that chooses optional independent hardening.	Yes	Yes Protocol-defined.	Pending immediately once grant is accepted; stronger after hardening starts.	After its own Bitcoin-backed maturity path.	Closest to root-like assurance, but reintroduces per-name Bitcoin cost.
alice after using alice@bitcoin	User later acquires a root name and migrates usage.	Yes	Yes	After winning root auction.	After root maturity.	Practical sovereignty upgrade, but it is a new root name, not the same subname.

Key distinction

A finalized subname can be self-sovereign against the root operator: the root cannot revoke or replace the subname owner's key. It is still namespace sovereignty, not identical to owning the root itself.

Resolver role

Resolvers verify roots, grants, owner-key update chains, and proofs. They provide data availability and witness infrastructure; they should not become registrars.